MOHAMED KHALID M JAFFAR

Email: khalid26@umd.edu | **Phone:** +1 240-733-7069 | **Portfolio:** khalid2696.github.io | **LinkedIn**.com/in/khalid-m-jaffar

PROFESSIONAL SUMMARY

Robotics and autonomy researcher with a decade of experience across diverse platforms, including ground, aerial, and legged robots, UAVs, and rockets. Expertise in motion planning, control, optimization, and learning-based methods for **safety-critical autonomous systems**, translating theory into real-world deployments. Focused on advancing reliable, intelligent autonomy through rigorous research, engineering, and system-level integration.

EDUCATION

University of Maryland, College Park, USA — **Expected Jun 2026**
Doctor of Philosophy (PhD) in Aerospace Engineering — **GPA: 3.62/4**

Indian Institute of Technology Madras, India — **2013 – 2018**
Dual Degree (B.Tech & M.Tech) in Aerospace Engineering — *Class Rank* **1**
Minor in Robotics — **GPA: 9.06/10**

SKILLS AND EXPERTISE

- **Planning & Control**: Trajectory optimization, Kinodynamic planning, MPC, LQR, Behavior planning, A*/RRT*
- **Navigation & Perception**: Kalman filters, EKF, Sensor fusion (IMU, LiDAR, RGB-D), Semantic segmentation
- **Machine Learning & AI**: Gaussian process, LLM/VLM integration for Physical AI, scikit-learn, PyTorch, CLIP
- **Languages & Tools**: Python, C++, MATLAB, C, ROS/ROS2, CasADi, CVXPY, Gurobi, Simulink, SolidWorks

RELEVANT WORK EXPERIENCE

Research Scientist Intern | Mitsubishi Electric Research Labs (MERL) — **2025–present**
*Project: Adaptive spatio-temporal monitoring using a heterogeneous mobile robot team — **US patent pending***

- Developed a multi-robot monitoring system using **12+** aerial and ground vehicles; implemented robust estimation and optimal task allocation while satisfying motion, battery, and collision-avoidance constraints
- Integrated data-driven learning with multi-agent coordination via Gaussian processes and integer programming
- Reduced monitoring effort by **85%** over non-learning baselines while enforcing hard uncertainty thresholds

Graduate Research Assistant | Motion and Teaming Lab, University of Maryland — **2018–25**
PhD Thesis: Real-time feedback motion re-planning of mobile robots in dynamic spaces

- Designed uncertainty-aware motion planning algorithms with formal Lyapunov-based safety guarantees for mobile robots in obstacle-cluttered environments, achieving robust replanning speeds of **50+ Hz**
- Released `computeSOSFunnels`, a MATLAB toolbox for invariant reachable set synthesis across **4** robot platforms
- Developed fault-tolerant control strategies for multirotor aerial vehicles with **up to 3** complete rotor failures

Visiting Researcher | U.S. Army Research Lab & GAMMA Lab, University of Maryland — **Summer 2024**
Project: Long-range navigation of legged robots in unstructured outdoor environments

- Field-deployed quadruped autonomous navigation (**>200 m**) in unmapped terrain with trees, tall grass, and mud
- Built multi-level costmaps from LiDAR and RGB camera sensor fusion for terrain-aware planning; used CLIP for semantic scene understanding and GPT-4o/Gemini for behavior-rule guided traversability estimation

Systems Lead | Airbrake subteam, Terrapin Rocket team | Spaceport America Cup — **2021–24**
Project: Real-time altitude regulation of a competition rocket via deployable aerodynamic braking flaps

- Engineered an airbrake avionics system, integrating sensors (IMU, barometer, magnetometer), MCU, and telemetry
- Implemented an MPC and EKF-based closed-loop control in C++, reducing apogee error from **2000 ft to 70 ft**
- Led a 6-student team over two years; placed **2nd in category** in 2023 and **1st overall** in 2024, out of **150 teams**

Master's Research | RAFT Lab & Robotics Lab, IIT Madras **2017–18**

Thesis: Onboard maneuver control of quadrotors in tight indoor spaces

- Benchmarked 4 nonlinear controllers; integral SMC outperformed alternatives in robustness against wind gusts
- Conducted **70+ hours** of hardware-in-the-loop tests for system identification and evaluation of control strategies

Principal Investigator | Innovative Projects Scheme | IC&SR, IIT Madras **Fall 2017**

Project: Autopilot flight controller board for multirotor UAVs

- Designed and fabricated a custom embedded autopilot board for real-time flight stabilization
- Achieved **<10 ms** end-to-end control latency through RTOS and assembly-level programming

Autonomy Lead | Team Noctua, DeTect Technologies | Tech startup **2015–17**

Product: Aerial robot operations for infrastructure inspection in oil and gas industrial plants

- Founding-stage member of a seed-funded startup, leading technology development and operations; the company has since grown to a **$138M valuation** (Series B)
- Developed localization and control algorithms for aerial robots to autonomously inspect pipelines and chimneys in petrochemical plants, reducing inspection time by **80%** compared to manual methods

Undergraduate Research Intern | IIT Madras & Defense R&D Organisation (DRDO) **Fall 2015**

Project: Autonomous target-tracking multirotor unmanned aerial vehicle (UAV)

- Implemented guidance & control strategies for a VTOL UAV to track reference trajectories with **0.2 m accuracy**

SELECTED PUBLICATIONS

Journal and Peer-Reviewed Conference Articles

- **M.K.M. Jaffar**, S. Di Cairano, A. Vinod. "Adaptive spatio-temporal monitoring using constrained mobile robots." *Conference on Decision and Control* (under review)
- **M.K.M. Jaffar** and M. Otte. "Online feedback motion re-planning in dynamic spaces using invariant funnels." *International Journal of Robotics Research* (preprint)
- K. Weerakoon, …, **M.K.M. Jaffar**, D. Manocha. "Behavioral rule guided autonomy using LLMs and VLMs for robot navigation in outdoor scenes." *ICRA* **(2025)**
- E. Bregin and **M.K.M. Jaffar**. "Development of a drag-modulating feedback system to control a rocket's ascent." *AIAA SciTech* **(2024)**
- **M.K.M. Jaffar** and M. Otte. "PiP-X: Funnel-based online feedback motion planning in dynamic environments." *Algorithmic Foundations of Robotics* [***best paper consideration***] **(2022)**

Open-source Software Releases

- **M.K.M. Jaffar**. "`computeSOSFunnels`: A MATLAB framework for invariant funnel synthesis using sum-of-squares optimization." *v1.0.0 - GitHub* **(2025)**

15+ peer-reviewed publications in ICRA, IROS, RA-L, CDC. A complete list is available on my **Google Scholar**.

AWARDS AND HONORS

- Ann G. Wylie **Dissertation Fellowship** for pursuing *6-month* academic research (Fall 2024)
- **Dean's Fellowship** from the University of Maryland, College Park (2019)
- **Gold Medalist** for **best overall academic record** in Dual Degree Aerospace Engineering (2018)
- **NTU-India Connect Scholarship** from NTU, Singapore; *1 among 20 students from India* (2016)

PROFESSIONAL SERVICE AND LEADERSHIP

- **Head Instructor, CAD Lab**: Taught **150+** undergraduate students core computer-aided design concepts through hands-on SolidWorks training; managed and supervised 4 teaching fellows at UMD College Park **Spring 2025**
- **Academic Chair** of Graduate Student Advisory Committee, Aerospace Department, UMD **2021–22**
- **Chief Strategist and Mentor** for student teams in **Robotics**, **Electronics**, and **Aero Clubs** at the Centre for Innovation (CFI), IIT Madras **2016–18**